

Data analysis and the representation of data

Collecting numbers is the easy part — this lesson is about turning them into a picture that tells the truth.

LESSON 79–80

2 × 40 MIN

ALGEBRA 8, CHAPTER IV §26

STAGE 9 · 6.1, 15.1, 15.5

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Collect data and organise it in a frequency table.
- Choose a suitable chart for a given kind of data.
- Draw and read a frequency polygon.
- Recognise a misleading graph.

TEXTBOOK

National	Chapter IV, pp. 165–172
Cambridge	Learner’s Book pp. 129–137, 318–330
Workbook	Workbook 6.1, 15.1

01

Explanation

From raw numbers to a table

A list of 40 test marks tells you nothing at a glance. A **frequency table** does.

MARK	0–10	10–20	20–30	30–40	40–50
Frequency	2	5	8	6	3

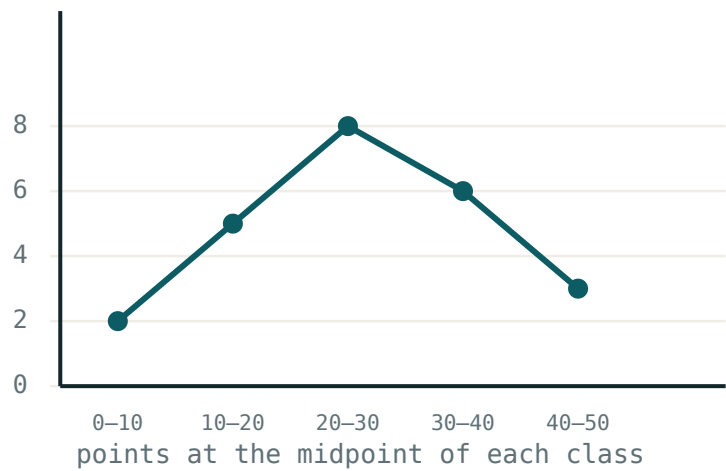
The groups are **classes**. Keep them equal in width, and say clearly which end belongs where — “10–20” usually means $10 \leq x < 20$.

SAMPLING

When you cannot ask everybody, you ask a **sample**. A good sample is large enough and chosen at random. Asking only your friends, or only people outside a sports hall, produces **bias** — the answer is wrong before any arithmetic starts.

Which chart?

CHART	USE IT FOR	WATCH OUT FOR
Bar chart	comparing separate categories	bars must be equal width
Pie chart	showing parts of one whole	useless for comparing two different totals
Line graph	change over time	only for continuous data
Frequency polygon	the shape of a grouped distribution	plot at the class midpoint



A frequency polygon: plot each frequency above the midpoint of its class, then join the points with straight lines.

The polygon shows the **shape** of the distribution — where the bulk of the data sits, and whether it leans to one side.

Misleading graphs

THREE TRICKS TO LOOK FOR

1. A vertical axis that does not start at zero — small differences look enormous.

2. Unequal class widths drawn as equal bars.

3. Pictures scaled in two directions — doubling the height and the width multiplies the apparent size by four, not two.

Ask of every graph: what is on each axis, where does the scale start, and does the picture match the numbers?

02 Worked examples

EXAMPLE 1 Twenty marks are: 5, 12, 18, 23, 25, 27, 29, 31, 33, 34, 36, 38, 41, 44, 8, 15, 22, 26, 30, 35. Make a frequency table with classes of width 10.

① 0–10: 5, 8 → 2

Tally each value into its class.

② 10–20: 12, 18, 15 → 3

③ 20–30: 23, 25, 27, 29, 22, 26 → 6

④ 30–40: 31, 33, 34, 36, 38, 30, 35 → 7

⑤ 40–50: 41, 44 → 2

Total $2 + 3 + 6 + 7 + 2 = 20$ ✓

ANSWER

2, 3, 6, 7, 2

EXAMPLE 2 Which chart suits “the number of learners choosing each of four sports”?

① Four separate categories, not a time series.

② A bar chart compares categories directly.

③ A pie chart also works if you want the share of the whole class.
But it hides the actual numbers.

ANSWER

A bar chart (or a pie chart for proportions).

EXAMPLE 3 A graph shows sales rising from 98 to 102 with the axis starting at 96. Why is it misleading?

- 1 The rise is 4 out of 98 — about 4%.
- 2 With the axis starting at 96, the bar for 102 looks three times the bar for 98.
- 3 Redrawing from zero shows the two bars almost equal.

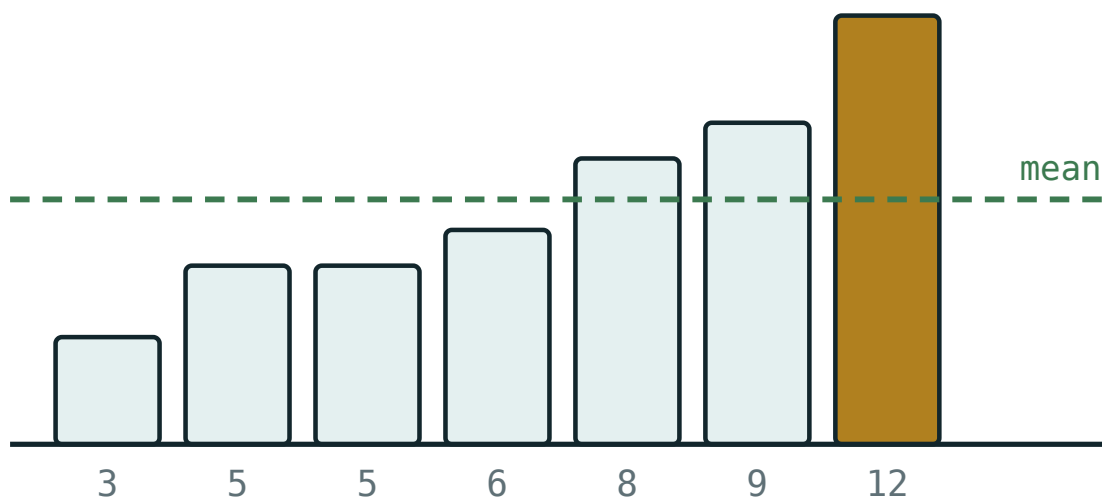
ANSWER

The truncated axis exaggerates a small change.

03 Interactive model

Change one value and watch which average moves — the mean always, the median sometimes, the mode rarely.

What the data looks like



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Data	Ma'lumotlar	Данные
Frequency	Chastota	Частота
Frequency table	Chastotalar jadvali	Таблица частот
Class (interval)	Sinf (oralig)	Класс (интервал)
Bar chart	Ustunli diagramma	Столбчатая диаграмма
Pie chart	Doiraviy diagramma	Круговая диаграмма
Frequency polygon	Chastotalar ko'pburchagi	Полигон частот
Sample	Tanlanma	Выборка
Bias	Nomutanosiblik	Смещение (необъективность)
Misleading graph	Chalg'ituvchi grafik	Вводящий в заблуждение график

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A frequency polygon is plotted at:

the class boundaries

the class midpoints

the largest value

zero

A pie chart is best for:

change over time

parts of one whole

comparing two different totals

raw data

Asking only your friends creates:

a large sample

bias

a frequency table

a fair sample

A vertical axis starting at 96 instead of 0:

is always wrong

exaggerates small differences

hides large differences

makes no difference

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *What is the frequency of a value?*

ANSWER How many times it occurs.

2. *Marks 2, 5, 5, 7, 5. Find the frequency of 5.*

ANSWER 3

3. *Which chart shows parts of a whole?*

ANSWER A pie chart.

4. *Which chart shows change over time?*

ANSWER A line graph.

5. *A class has 8, 12, 15, 5 learners in four groups. Find the total.*

ANSWER 40

6. *In a frequency polygon, where is each point plotted?*

ANSWER Above the midpoint of the class.

7. *Class 20–30. Find its midpoint.*

ANSWER 25

Medium · the standard question

7 PROBLEMS

1. Marks 5, 12, 18, 23, 25, 27, 29, 31. Make classes of width 10 and give the frequencies.

ANSWER 0–10: 1, 10–20: 2, 20–30: 4, 30–40: 1

2. Frequencies 3, 7, 10, 6, 4. Find the total and the modal class.

ANSWER 30; the third class

3. Which chart for “learners choosing each of four sports”?

ANSWER A bar chart.

4. A pie chart has 360° . What angle represents 25%?

ANSWER 90°

5. A pie chart has 40 learners. What angle represents 10 of them?

ANSWER 90°

6. Why is asking people outside a gym about exercise biased?

ANSWER They are far more likely to exercise than the general population.

7. Class midpoints 5, 15, 25, 35 with frequencies 2, 5, 8, 3. Which class is modal?

ANSWER 20–30

Hard · stretch and reason

7 PROBLEMS

1. *A graph shows sales rising 98 → 102 with the axis from 96. Explain why it misleads.*

ANSWER A 4% rise is drawn to look like a threefold one; the axis should start at 0.

2. *A pie chart of 90 learners: one sector is 120°. How many learners is that?*

ANSWER $\frac{120}{360} \times 90 = 30$

3. *Frequencies 4, 9, 12, 7, 3 for classes of width 5 starting at 0. Draw the frequency polygon points.*

ANSWER (2.5; 4), (7.5; 9), (12.5; 12), (17.5; 7), (22.5; 3)

4. *A survey of 20 people finds 15 like a product. Is that enough to claim 75% of the town does?*

ANSWER No — the sample is small and may not be random; the margin of uncertainty is large.

5. *Two pie charts show 60% for one school of 100 and 60% for another of 1000. Are the sectors comparable?*

ANSWER The angles are equal, but they represent 60 and 600 people — a pie chart hides the totals.

6. *Explain why doubling both the width and the height of a picture symbol misleads.*

ANSWER The area becomes four times as large, so the eye reads a fourfold increase.

7. *Design a fair way to sample 50 learners from a school of 800.*

ANSWER Number everyone 1–800 and choose 50 at random, or take a proportional number from each year group.

07 Homework

Homework — 5 problems

Algebra 8, Chapter IV, pp. 165–172, and Cambridge Learner's Book 6.1.

- Marks 3, 9, 14, 16, 21, 24, 25, 28, 31, 37. Make a frequency table with classes of width 10.
- Draw the frequency polygon for that table.

3. *A pie chart of 72 learners has a sector of 150° . How many learners is that?*
4. Name a suitable chart for each: monthly temperature; favourite subject; share of a budget.
5. Describe one way a graph in a newspaper could mislead, and how to check for it.

Mean, mode and median

Three different answers to “what is a typical value?” — and the reason a country can have a rising average income and a falling typical one.

LESSON 81–82

2 × 40 MIN

ALGEBRA 8, CHAPTER IV §27

STAGE 9 · 15.4

LESSON CLOCK

10'

24'

16'

26'

4'

Warm-up	10 min
Explanation	24 min
Interactive	16 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Calculate the mean, the mode and the median of a list.
- Find the range.
- Estimate the mean of grouped data using midpoints.
- Choose the most suitable average for a given set.

TEXTBOOK

National	Chapter IV, pp. 173–180
Cambridge	Learner’s Book pp. 331–338
Workbook	Workbook 15.4

01

Explanation

The three averages

DEFINITIONS

- Mean** — add everything and divide by how many: $\bar{x} = \frac{\text{sum of values}}{\text{number of values}}$.
- Mode** — the value that occurs most often. There may be none, or several.
- Median** — sort the values; the middle one. With an even count, the mean of the two middle ones.

And the **range** = largest – smallest. It is not an average; it measures **spread**.

For 3, 5, 5, 6, 8, 9, 12: mean $\frac{48}{7} \approx 6.9$, mode 5, median 6, range 9.

SORT FIRST, EVERY TIME

The median of 7, 2, 9 is 7, not 2. Writing the list in order takes five seconds and prevents the commonest error in the topic.

Which average to use

AVERAGE	GOOD BECAUSE	BAD BECAUSE
Mean	uses every value	one extreme value drags it a long way
Median	ignores extremes	ignores most of the data
Mode	the only one for categories	may not exist, or may be at one end

Nine people earn 2 million and one earns 100 million. The **mean** is 11.8 million — a figure nobody earns. The **median** is 2 million, which describes the group honestly. A single **outlier** is exactly what the median is for.

Grouped data

Once the raw values are lost in classes, the mean can only be **estimated**. Assume everything in a class sits at its **midpoint**:

$$estimated\ mean = \frac{\sum (midpoint \times frequency)}{\sum frequency}$$

CLASS	MIDPOINT	F	MIDPOINT × F
0–10	5	2	10
10–20	15	5	75
20–30	25	8	200
30–40	35	6	210
40–50	45	3	135
Total		24	630

$$estimated\ mean = \frac{630}{24} = 26.25$$

The **modal class** is the one with the highest frequency — here 20–30.

02 Worked examples

EXAMPLE 1 Find the mean, mode, median and range of 4, 7, 3, 7, 9, 2, 7.

- ① sorted: 2, 3, 4, 7, 7, 7, 9
Sort first.

② $mean = \frac{39}{7} \approx 5.6$

- ③ $mode = 7$
It appears three times.

- ④ $median = 7$
The fourth of seven values.

⑤ $range = 9 - 2 = 7$

ANSWER

$mean \approx 5.6$, mode 7, median 7, range 7

EXAMPLE 2 Find the median of 12, 5, 8, 20, 3, 11.

- ① sorted: 3, 5, 8, 11, 12, 20
Six values — even.

- ② The two middle ones are 8 and 11.

③ $median = \frac{8 + 11}{2} = 9.5$
Their mean.

ANSWER

9.5

EXAMPLE 3 Estimate the mean for classes 0–10, 10–20, 20–30 with frequencies 4, 10, 6.

① midpoints 5, 15, 25

② $5 \cdot 4 + 15 \cdot 10 + 25 \cdot 6 = 20 + 150 + 150 = 320$

③ total frequency = 20

④ $\frac{320}{20} = 16$

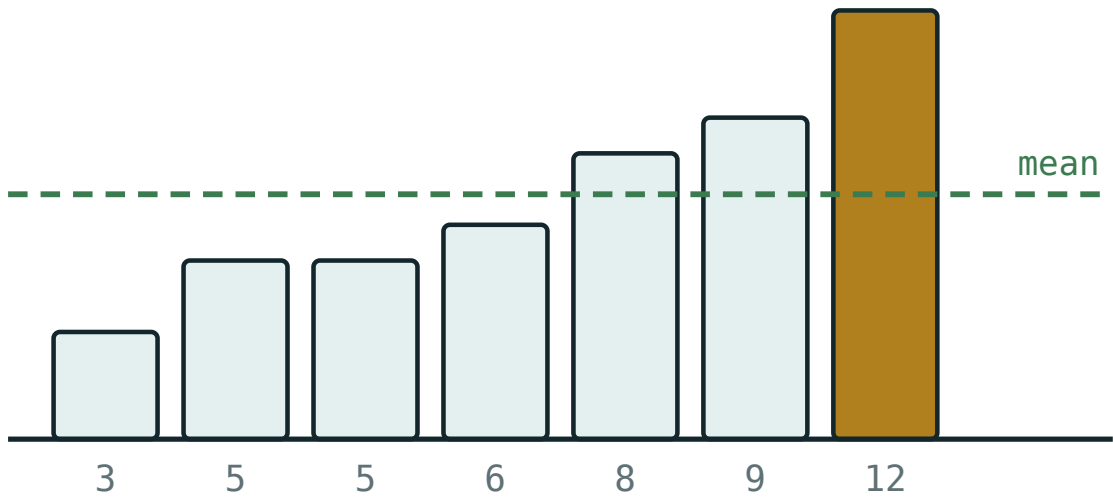
ANSWER

≈ 16

03 Interactive model

Push one value up to 14 and watch the mean move while the median stays still.

Mean, median and mode



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mean (average)	O'rtacha arifmetik	Среднее арифметическое
Mode	Moda	Мода
Median	Mediana	Медиана
Range	Kenglik (farq)	Размах
Sorted order	O'sish tartibida	В порядке возрастания
Outlier	Chetlangan qiymat	Выброс
Grouped data	Guruhlangan ma'lumotlar	Сгруппированные данные
Midpoint of a class	Sinf o'rtasi	Середина интервала
Estimate	Baholash	Оценить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The median of 7, 2, 9 is:

One very large value most affects the:

For grouped data the mean is estimated using:

The range of 3, 5, 5, 12 is:

5

9

12

6.25

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the mean of 4, 6, 8

ANSWER 6

2. Find the mode of 2, 3, 3, 5

ANSWER 3

3. Find the median of 1, 4, 9

ANSWER 4

4. Find the range of 2, 7, 11

ANSWER 9

5. Find the mean of 10, 20, 30, 40

ANSWER 25

6. Find the median of 3, 5, 7, 9

ANSWER 6

7. Find the mode of 1, 2, 3, 4

ANSWER None.

Medium · the standard question

7 PROBLEMS

1. Find the mean, mode, median and range of 4, 7, 3, 7, 9, 2, 7

ANSWER $\approx 5.6, 7, 7, 7$

2. Find the median of 12, 5, 8, 20, 3, 11

ANSWER 9.5

3. The mean of five numbers is 12. Find their total.

ANSWER 60

4. Four numbers have mean 9. A fifth is added and the mean becomes 10. Find it.

ANSWER $50 - 36 = 14$

5. Estimate the mean for classes 0–10, 10–20, 20–30 with frequencies 4, 10, 6.

ANSWER 16

6. Find the modal class for frequencies 3, 9, 12, 5.

ANSWER The third class.

7. Nine people earn 2 and one earns 100 (millions). Find the mean and median.

ANSWER mean = 11.8, median = 2

Hard · stretch and reason

7 PROBLEMS

1. *The mean of 6 numbers is 15. One number, 20, is removed. Find the new mean.*

ANSWER $\frac{90 - 20}{5} = 14$

2. *A set of 5 numbers has mean 8, median 7 and mode 6. Give a possible set.*

ANSWER 6, 6, 7, 10, 13 — mean 8.4; adjust to 6, 6, 7, 9, 12 (mean 8) ✓

3. *Estimate the mean for classes 0–20, 20–40, 40–60 with frequencies 5, 12, 3.*

ANSWER $\frac{10 \cdot 5 + 30 \cdot 12 + 50 \cdot 3}{20} = \frac{560}{20} = 28$

4. *Why is the median better than the mean for house prices?*

ANSWER A few very expensive houses drag the mean far above what a typical house costs.

5. *The mean of 10 numbers is 6. Adding an eleventh makes the mean 6.5. Find it.*

ANSWER $71.5 - 60 = 11.5$

6. *A test has mean 60 for 20 boys and 70 for 30 girls. Find the mean for all 50.*

ANSWER $\frac{1200 + 2100}{50} = 66$

7. *Can the mean, median and mode all be different? Give an example.*

ANSWER Yes — 1, 2, 2, 5, 10: mean 4, median 2, mode 2. Try 1, 2, 2, 6, 14: mean 5, median 2, mode 2.

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 173–180. Sort before finding a median.

- Find the mean, mode, median and range of 8, 3, 5, 8, 11, 8, 2
- Find the median of 14, 9, 21, 6, 17, 11
- The mean of seven numbers is 11. Find their total.

4. *Estimate the mean for classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 8, 6, 3.*
5. *Six numbers have mean 14. A seventh makes the mean 15. Find it.*
6. Explain in two sentences when the median is a better average than the mean.

Elements of combinatorics

— counting by listing

Before any formula, learn to list without repeating and without missing anything.

LESSON 83–85

3 × 40 MIN

ALGEBRA 8, CHAPTER IV §27

STAGE 9 · 12.1

LESSON CLOCK

18'

36'

30'

24'

12'

Warm-up: how many two-digit numbers use only 1, 2, 3?

18 min

Systematic listing

36 min

Tables of outcomes

30 min

Tree diagrams

24 min

Homework

12 min

LEARNING OBJECTIVES

- List every outcome of a small experiment in a systematic order.
- Use a table of outcomes for two stages.
- Draw a tree diagram and read the outcomes off its branches.
- Decide whether order matters in a given counting question.

TEXTBOOK

National

Chapter IV, pp. 181–188

Cambridge

Learner’s Book pp. 255–260

Workbook

Workbook 12.1

01

Explanation

Listing in an order you can trust

Combinatorics is the art of counting without counting one by one. But every method starts from the same discipline: **list in a fixed order**, so that nothing is written twice and nothing is forgotten.

How many two-digit numbers can be made from the digits 1, 2, 3 if a digit may repeat? Fix the first digit, then run through the second:

11, 12, 13 | 21, 22, 23 | 31, 32, 33

Nine numbers — three blocks of three. The blocks are what make the list trustworthy: as soon as you are working alphabetically, or from smallest to largest, a missing item is visible.

THE RULE OF THE LIST

Choose one thing to vary **last**. Hold everything else still while it runs through all its values, then step the next thing on by one.

Does order matter?

Two questions that sound the same are not the same:

QUESTION	ORDER?	FROM {A, B, C} CHOOSE 2	HOW MANY
Who is captain and who is vice-captain?	matters	<i>AB, BA, AC, CA, BC, CB</i>	6
Which two go on the trip?	does not matter	<i>AB, AC, BC</i>	3

When order does not matter, *AB* and *BA* are one and the same choice, so the count is halved. Always ask the question in words before you start listing.

ALSO ASK: MAY THINGS REPEAT?

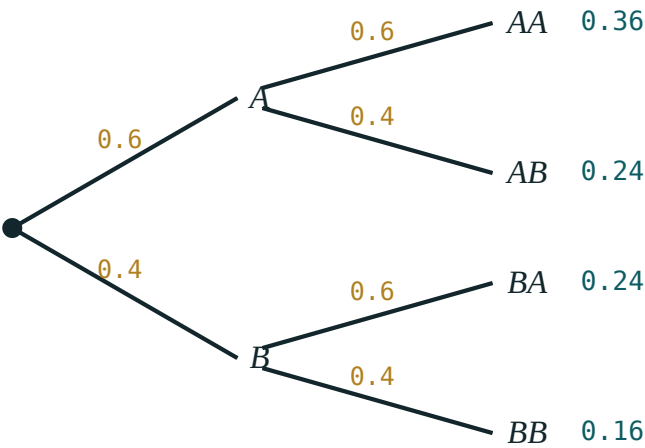
Two dice can both show 4 (repetition allowed). Two different pupils cannot be the same pupil (repetition not allowed). The answer changes the list.

Tables and tree diagrams

For an experiment in **two** stages, a table lists every outcome at once. Two dice:

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Thirty-six squares, so thirty-six equally likely outcomes. A sum of 7 fills a whole diagonal — six squares — which is why 7 is the commonest total.



multiply along a branch · add down the ends

A tree diagram. Each path from the root to a tip is one outcome; count the tips to count the outcomes.

A **tree diagram** handles three stages and more, and it keeps the order visible. Read one outcome by walking a single path from left to right.

02 Worked examples

EXAMPLE
1

A café offers 3 soups and 4 main dishes. List, systematically, the ways to choose one soup and one main.

1

$S_1M_1, S_1M_2, S_1M_3, S_1M_4$
Hold the soup still; run the main through all four values.

2

$S_2M_1, S_2M_2, S_2M_3, S_2M_4$
Step the soup on by one.

3

$S_3M_1, S_3M_2, S_3M_3, S_3M_4$
And again.

4

$3 \text{ blocks} \times 4 = 12$
The blocks are equal, so the total is a product.

ANSWER

12 meals

EXAMPLE 2 From the letters A, B, C, D, how many two-letter “words” can be made with no letter repeated?

① *AB, AC, AD*

First letter A — three choices left.

② *BA, BC, BD*

First letter B.

③ *CA, CB, CD, DA, DB, DC*

C and D likewise.

④ $4 \times 3 = 12$

Four first letters, three second letters each.

ANSWER

12 words

EXAMPLE 3 A coin is tossed three times. List all the outcomes and count those with exactly two heads.

① *HHH, HHT, HTH, HTT*

First toss H; the other two run through all four possibilities.

② *THH, THT, TTH, TTT*

First toss T.

③ *8 outcomes in total*

$2 \times 2 \times 2 = 8$ — three stages, two ways each.

④ *HHT, HTH, THH*

Pick out the ones with exactly two H.

ANSWER

8 outcomes; 3 of them have exactly two heads

03 Interactive model

Reveal one counting task at a time and let the class finish the list before you show the next step.

Count it by listing

Digits 1, 2, 3, 4. Two-digit numbers, digits **different**.

① 12, 13, 14

First
digit
1.

② 21, 23, 24

First
digit
2.

③ 31, 32, 34

First
digit
3.

④ 41, 42, 43

First
digit
4 —
four
blocks
of
three.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Outcome	Natija	Исход
Sample space	Elementar natijalar fazosi	Пространство исходов
Systematic listing	Tartibli sanab chiqish	Систематический перебор
Tree diagram	Daraxt diagrammasi	Дерево вариантов
Table of outcomes	Natijalar jadvali	Таблица исходов
Arrangement (order matters)	O'rin almashtirish	Размещение (порядок важен)
Selection (order does not matter)	Tanlash	Сочетание (порядок не важен)
Repetition allowed	Takrorlanish bilan	С повторением
Exhaustive list	To'liq ro'yxat	Полный перечень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

How many two-digit numbers use only the digits 5, 6, 7, repeats allowed?

From 5 people, how many ways can a captain and a vice-captain be chosen?

Two dice are thrown. How many outcomes are there altogether?

In which question does order NOT matter?

Who wins gold and who wins silver?

Which two books do I borrow?

What is the PIN code?

Who sits first and who sits second?

A tree diagram for three stages with 2 branches each has how many tips?

6

8

9

12

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *List all two-digit numbers formed from 1, 2 with repeats allowed.*

ANSWER 11, 12, 21, 22

2. *List all two-letter words from A, B, C with no repeats.*

ANSWER AB, AC, BA, BC, CA, CB

3. *How many outcomes when a coin is tossed twice?*

ANSWER 4

4. *List the ways to choose 2 fruits from apple, pear, plum.*

ANSWER apple–pear, apple–plum, pear–plum

5. *How many outcomes when a die is thrown and a coin tossed?*

ANSWER 12

6. *List all three-digit numbers using each of 1, 2, 3 once.*

ANSWER 123, 132, 213, 231, 312, 321

7. *Does order matter in “choose 3 pupils for the choir”?*

ANSWER No.

Medium · the standard question

7 PROBLEMS

1. *How many two-digit numbers from 0, 1, 2, 3 (no repeats, no leading 0)?*

ANSWER $3 \times 3 = 9$

2. *A menu has 2 starters, 3 mains, 2 desserts. How many meals?*

ANSWER 12

3. *Two dice are thrown. How many outcomes give a total of 8?*

ANSWER 5 — (2,6),(3,5),(4,4),(5,3),(6,2)

4. *From 6 people, how many ways to pick a president and a secretary?*

ANSWER $6 \times 5 = 30$

5. *From 6 people, how many ways to pick a committee of 2?*

ANSWER $30 \div 2 = 15$

6. *A coin is tossed four times. How many outcomes?*

ANSWER $2^4 = 16$

7. *How many outcomes of two dice have both numbers the same?*

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. *How many three-digit numbers use only 1, 2, 3 and contain at least one 3?*

ANSWER $27 - 8 = 19$

2. *How many two-digit even numbers can be made from 1, 2, 3, 4, 5 with no repeats?*

ANSWER $2 \times 4 = 8$

3. *Two dice are thrown. In how many outcomes is the total a prime number?*

ANSWER 15 — totals 2, 3, 5, 7, 11 give $1 + 2 + 4 + 6 + 2$

4. *From 5 pupils, how many ways to seat 3 of them in a row?*

ANSWER $5 \times 4 \times 3 = 60$

5. *How many four-letter “words” from A, B with repeats, having exactly two A’s?*

ANSWER 6 — AABB, ABAB, ABBA, BAAB, BABA, BBAA

6. *A PIN uses 4 digits from 0–9, repeats allowed. How many begin with an odd digit?*

ANSWER $5 \times 10^3 = 5000$

7. *Explain why choosing 2 from n gives half of $n(n - 1)$.*

ANSWER Each unordered pair is counted twice among the $n(n - 1)$ ordered pairs — once as XY and once as YX .

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 181–188. Show your list, not just the number.

1. List all two-digit numbers made from 2, 4, 6 with repeats allowed.
2. From A, B, C, D choose 2 with order mattering — list them all.
3. A shop sells 4 shirts and 3 ties. How many shirt-and-tie sets?
4. Two dice are thrown. List the outcomes with a total of 9.
5. A coin is tossed three times. List the outcomes with at most one head.

6. Write one counting question where order matters and one where it does not.

The basic principle of counting

When each stage is independent, multiply. That single sentence replaces a page of listing.

LESSON 86–88

3 × 40 MIN

ALGEBRA 8, CHAPTER IV §28

STAGE 9 · 12.2–12.3

LESSON CLOCK

18'

36'

24'

30'

12'

Recall yesterday's lists	18 min
The multiplication rule	36 min
Arrangements and factorials	24 min
Addition rule and "at least one"	30 min
Homework	12 min

LEARNING OBJECTIVES

- State and apply the multiplication principle of counting.
- Count arrangements of n different objects.
- Use the addition principle for cases that cannot happen together.
- Count by subtracting from the total when "at least one" appears.

TEXTBOOK

National	Chapter IV, pp. 189–197
Cambridge	Learner's Book pp. 261–270
Workbook	Workbook 12.2–12.3

01

Explanation

The multiplication rule

BASIC PRINCIPLE OF COUNTING

If a task is done in k stages, the first in n_1 ways, the second in n_2 ways and so on — **whatever was chosen before** — then the whole task can be done in

$$n_1 \cdot n_2 \cdot \dots \cdot n_k$$

ways.

Yesterday's café gave $3 \times 4 = 12$ meals, and the reason is now visible in the list: three equal blocks of four. The rule only replaces the listing — it does not change the answer.

THE CONDITION THAT MATTERS

The number of ways at each stage must not depend on *which* choice was made earlier. “Four shirts, three ties” multiplies. “Four shirts, and the number of matching ties depends on the shirt” does not.

Arrangements and the factorial

In how many orders can 5 different books stand on a shelf? Five choices for the first place, then four books remain, then three, two, one:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

This product has a name: $5! = 120$, read “five factorial”. In general

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1, \text{ and by agreement } 0! = 1$$

N	1	2	3	4	5	6	7
$n!$	1	2	6	24	120	720	5040

Factorials grow terrifyingly fast — $10! = 3\,628\,800$. That is exactly why counting by listing stops being practical and the rule takes over.

If only r of the n objects are placed, stop the product after r factors: seating 3 of 5 pupils gives $5 \cdot 4 \cdot 3 = 60$.

Adding, and counting the opposite

ADDITION RULE

If two cases **cannot happen together**, count each and add. Going to Samarkand by one of 3 trains **or** one of 5 buses gives $3 + 5 = 8$ ways — not 15, because you take one vehicle, not both.

Multiply for “and then”; **add** for “or”. Reading the word in the question is half the work.

“At least one” is almost always easier **backwards**. How many three-digit codes from 1, 2, 3 contain at least one 3?

$$\text{all codes } 3^3 = 27 - \text{codes with no 3 } 2^3 = 8 = 19$$

Counting the opposite and subtracting is called **complementary counting**. Use it whenever the direct count splits into many cases.

02 Worked examples

EXAMPLE 1 A car number plate has 2 letters (from 26) followed by 3 digits. How many plates are possible?

- ① 26 ways for the first letter
Letters may repeat unless the question forbids it.
- ② 26 ways for the second
- ③ $10 \cdot 10 \cdot 10$ for the digits
Three independent stages.
- ④ $26 \cdot 26 \cdot 10^3 = 676\,000$
Multiply every stage together.

ANSWER

676 000

EXAMPLE 2 In how many ways can 6 pupils stand in a queue, if two particular pupils must stand together?

- ① Treat the pair as one block
Now there are 5 items to arrange.
- ② $5! = 120$
Arrangements of the block and the other four.
- ③ $\times 2$
The two inside the block can swap.
- ④ $120 \cdot 2 = 240$

ANSWER

240 ways

EXAMPLE 3 How many four-digit numbers have all digits different and are odd?

① *last digit: 5 ways*

It must be one of 1, 3, 5, 7, 9 — deal with the strictest condition first.

② *first digit: 8 ways*

Not 0, and not the digit already used at the end.

③ *second digit: 8 ways*

Ten digits less the two used.

④ *third digit: 7 ways*

⑤ $5 \cdot 8 \cdot 8 \cdot 7 = 2240$

ANSWER

2240 numbers

03 Interactive model

Work through the counting tasks one stage at a time; ask the class for the next factor before revealing it.

Multiply, add, or subtract?

26 letters, 10 digits, repetition allowed. How many passwords?

① $26 \cdot 26 \cdot 26$

Three
letter
stages,
independent.

② $\times 10 \cdot 10$

Two
digit
stages.

③ $17\,576 \cdot 100$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Independent choices	Bog'liq bo'lmagan tanlovlar	Независимые выборы
Factorial	Faktorial	Факториал
Permutation	O'rin almashtirish	Перестановка
Complementary counting	To'ldiruvchi hisoblash	Подсчёт через дополнение
Stage (step)	Bosqich	Этап
At least one	Kamida bitta	Хотя бы один
Total number of ways	Usullar soni	Число способов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A shop has 5 jackets and 4 hats. How many jacket-and-hat outfits?

What is $6!$?

You may travel by one of 4 trains or one of 6 buses. How many ways?

How many three-digit codes from 1, 2, 3, 4 have no digit repeated?

64

24

12

48

How many of the $2^5 = 32$ outcomes of five coin tosses have at least one head?

31

16

32

30

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Find $4!$*

ANSWER 24

2. *3 shirts, 5 ties — how many sets?*

ANSWER 15

3. *How many two-digit codes from 0–9, repeats allowed?*

ANSWER 100

4. *In how many orders can 3 books stand on a shelf?*

ANSWER 6

5. *2 roads from A to B and 3 from B to C. How many routes A to C?*

ANSWER 6

6. *A menu has 4 mains or 3 salads. Choose one dish — how many ways?*

ANSWER 7

7. *Find $5! \div 3!$*

ANSWER 20

Medium · the standard question

7 PROBLEMS

1. *How many three-digit numbers have all digits different?*

ANSWER $9 \cdot 9 \cdot 8 = 648$

2. *In how many ways can 5 pupils be seated in a row?*

ANSWER 120

3. *A code is 2 letters (26) then 4 digits. How many codes?*

ANSWER $26^2 \cdot 10^4 = 6\,760\,000$

4. *How many four-digit even numbers use digits 1–5 with repeats?*

ANSWER $5^3 \cdot 2 = 250$

5. *From 7 pupils, how many ways to choose a captain, then a vice-captain?*

ANSWER 42

6. *How many three-letter words from A–E have no repeated letter?*

ANSWER 60

7. *How many of the 36 two-dice outcomes contain at least one 6?*

ANSWER $36 - 25 = 11$

Hard · stretch and reason

7 PROBLEMS

1. *In how many ways can 5 people sit in a row if two of them must sit together?*

ANSWER $4! \cdot 2 = 48$

2. *In how many ways can 5 people sit in a row if two of them must NOT sit together?*

ANSWER $120 - 48 = 72$

3. *How many four-digit numbers with distinct digits are divisible by 5?*

ANSWER 952 — ending 0: $9 \cdot 8 \cdot 7 = 504$; ending 5: $8 \cdot 8 \cdot 7 = 448$

4. *How many three-digit numbers from 1–6 (repeats allowed) contain at least two 3s?*

ANSWER 16 — exactly two: $3 \cdot 5 = 15$, all three: 1

5. *In how many ways can the letters of MATH be arranged so that M comes first?*

ANSWER $3! = 6$

6. *A committee of a chair and 2 ordinary members is chosen from 6 people. How many ways?*

ANSWER $6 \cdot 10 = 60$ — chair 6 ways, then 2 from 5 gives 10

7. *Show that the number of subsets of a set with n elements is 2^n .*

ANSWER Each element is either in or out — n independent two-way choices, so 2^n by the multiplication rule.

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 189–197. Say which rule you used in each answer.

- Find $6!$ and $7!$
- A café has 3 soups, 5 mains and 2 desserts. How many three-course meals?
- How many three-digit numbers have all digits different?
- In how many orders can 4 photographs be hung in a row?
- How many three-digit codes from 1–5 contain at least one 1?

6. In one sentence, explain when you multiply and when you add.

Control work 7 · Data and counting

The Chapter IV assessment, and the four errors it produces every year.

LESSON 89–90

2 × 40 MIN

ALGEBRA 8, CHAPTER IV

STAGE 9 · 12.1–12.3, 15.1

LESSON CLOCK

4'

72'

4'

Setting up	4 min
The paper	72 min
Collect in	4 min

LEARNING OBJECTIVES

- Assess averages, grouped data, charts, listing and the counting principle.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of Chapter IV
Cambridge	Learner's Book pp. 255–270, 318–324
Workbook	Workbook 12.1–12.3

01

Explanation

Lesson 89 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · mean, mode, median and range of a small list
- task 2 · read a value from a frequency table
- task 3 · estimate the mean of grouped data
- task 4 · choose and justify a suitable chart
- task 5 · list all outcomes systematically
- task 6 · the multiplication rule
- task 7 · an “at least one” count

Marking note: task 1 loses a mark if the list is not sorted before the median is read.

Lesson 90 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

- Median read from an unsorted list.** Sorting is part of the method, not a nicety.
- Grouped mean without midpoints** — adding the class boundaries, or the frequencies alone.

3. **Adding when the question says “and then”** — the counting rules swapped over.
4. **“At least one” counted directly**, producing double-counted cases instead of *total* — *none*.

Each pupil writes the error number beside every lost mark, then re-solves that task in full.

02 Worked examples

EXAMPLE 1 Find the error: the median of 7, 2, 9, 4, 5 is 9

- 1 The list was never sorted.
The middle **of the sorted list** is what the median means.
- 2 2, 4, 5, 7, 9
Sort first.
- 3 $median = 5$
The third of five values.

ANSWER

5

EXAMPLE 2 Find the error: classes 0–10, 10–20 with frequencies 4, 6 give a mean of $(10 + 20) \div 2 = 15$

① Class **midpoints** are needed, weighted by frequency.

② *midpoints 5 and 15*

③
$$\frac{5 \cdot 4 + 15 \cdot 6}{10} = \frac{110}{10} = 11$$

ANSWER

11

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: the median of 12, 5, 8, 3, 20, 7 is 8 and 3 $\rightarrow 5.5$

① 3, 5, 7, 8, 12, 20

Sort
the
list
first
—
always.

② Six values, so the median is the mean of the 3rd and 4th.

③ $(7 + 8) \div 2$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Mean	O'rta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Modal class	Moda sinfi	Модальный класс
Estimated mean	Taxminiy o'rta	Оценка среднего
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Complementary counting	To'ldiruvchi hisoblash	Подсчёт через дополнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The median of 7, 2, 9, 4, 5 is:

For grouped data the estimated mean uses:

3 starters and 4 mains give how many two-course meals?

The best way to count “at least one 6 in two dice throws” is:

add the cases

$36 - 25$

$6 + 6$

$2 \cdot 6$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Find the mean, mode, median and range of 4, 7, 4, 9, 6

ANSWER mean 6, mode 4, median 6, range 5

2. **Task 2.** Marks 1–5 have frequencies 2, 5, 8, 3, 2. How many pupils?

ANSWER 20

3. **Task 3.** Estimate the mean for 0–10, 10–20, 20–30 with frequencies 3, 5, 2.

ANSWER 14

4. **Task 4.** Which chart shows parts of one whole?

ANSWER A pie chart.

5. **Task 5.** List all two-digit numbers from 1, 2, 3 with repeats.

ANSWER 11, 12, 13, 21, 22, 23, 31, 32, 33

6. **Task 6.** 5 hats and 3 scarves — how many sets?

ANSWER 15

7. **Task 7.** How many of the 8 outcomes of three coin tosses have at least one head?

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Find the mean, mode, median and range of 12, 5, 12, 8, 3

ANSWER mean 8, mode 12, median 8, range 9

2. **Task 2.** Frequencies 4, 6, 7, 3 for scores 1–4. Find the modal score.

ANSWER 3

3. **Task 3.** Estimate the mean for 0–20, 20–40, 40–60 with frequencies 6, 10, 4.

ANSWER 26

4. **Task 4.** Why can a graph starting at 90 rather than 0 mislead?

ANSWER Small differences are stretched so they look large.

5. **Task 5.** List the ways to choose 2 pupils from A, B, C, D.

ANSWER AB, AC, AD, BC, BD, CD

6. **Task 6.** A code has 2 letters (26) then 2 digits. How many codes?

ANSWER $26^2 \cdot 100 = 67\,600$

7. **Task 7.** How many three-digit codes from 1–5 contain at least one 2?

ANSWER $125 - 64 = 61$

Hard · stretch and reason

7 PROBLEMS

1. *The mean of 8 numbers is 12. Two numbers, 5 and 19, are removed. Find the new mean.*

ANSWER $\frac{96 - 24}{6} = 12$

2. *A set of five whole numbers has mean 7, median 6 and mode 5. Find one such set.*

ANSWER 5, 5, 6, 9, 10

3. *Estimate the mean for 0–5, 5–10, 10–15, 15–20 with frequencies 2, 7, 8, 3.*

ANSWER $\frac{5 + 52.5 + 100 + 52.5}{20} = 10.5$

4. *How many four-digit numbers have all digits different and start with an odd digit?*

ANSWER $5 \cdot 9 \cdot 8 \cdot 7 = 2520$

5. *From 8 people, how many ways to pick a chair and a secretary?*

ANSWER 56

6. *How many three-digit codes from 1–6 have at least two equal digits?*

ANSWER $216 - 120 = 96$

7. *Explain why the mean of a grouped table is only an estimate.*

ANSWER Every value in a class is replaced by the midpoint, so the individual values are lost and the total is only approximated.

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.
2. Beside each, write the error number (1–4) that caused it.
3. Find the mean, mode, median and range of 6, 9, 6, 14, 5
4. Estimate the mean for 0–10, 10–20, 20–30, 30–40 with frequencies 2, 7, 8, 3.

5. *How many three-digit codes from 1–4 contain at least one 1?*

Revision · Sequences, the nth term and functions

Cambridge insert: the rule behind a pattern, written once and used for every term.

LESSON 91–93

3 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 9.1–9.3

LESSON CLOCK

18'

36'

30'

24'

12'

Continue three patterns	18 min
The nth term	36 min
Quadratic and special sequences	30 min
Function notation	24 min
Homework	12 min

LEARNING OBJECTIVES

- Continue a sequence and describe its term-to-term rule.
- Find the nth term of a linear sequence.
- Recognise square, cube and simple quadratic sequences.
- Use function notation and complete a table of values.

TEXTBOOK

National	Annual revision
Cambridge	Learner’s Book pp. 194–210
Workbook	Workbook 9.1–9.3

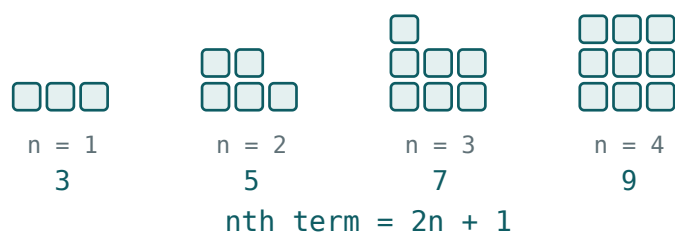
01

Explanation

Two ways to describe a sequence

Look at 5, 8, 11, 14, 17, ...

RULE	SAYS	GOOD FOR
term-to-term	“add 3 to the term before”	the next term
position-to-term (nth term)	$3n + 2$	the 100th term, straight away



A linear sequence grows by the same amount each step. The common difference is the coefficient of n in the n th term.

NTH TERM OF A LINEAR SEQUENCE

$n\text{th term} = dn + (a_1 - d)$

where d is the common difference and a_1 the first term. In practice: write $d \cdot n$, then adjust by whatever is needed to make $n = 1$ come out right.

For 5, 8, 11, ... the difference is 3, so start with $3n$: that gives 3, 6, 9, ... — every term is 2 too small. The rule is $3n + 2$. Check at $n = 4$: $12 + 2 = 14$ ✓.

Sequences that are not linear

SEQUENCE	PATTERN	NTH TERM
1, 4, 9, 16, 25	squares	n^2
1, 8, 27, 64	cubes	n^3
2, 4, 8, 16, 32	doubling	2^n
3, 6, 11, 18, 27	squares shifted	$n^2 + 2$
1, 3, 6, 10, 15	triangular	$\frac{1}{2}n(n + 1)$

The test is the **second difference**. If the differences themselves change by a constant, the sequence is quadratic:

$3, 6, 11, 18, 27 \rightarrow$ first differences $3, 5, 7, 9 \rightarrow$ second differences $2, 2, 2$

A constant second difference of 2 means an n^2 sits at the heart of the rule.

Function notation

A **function** is a machine: one input goes in, exactly one output comes out. Writing $f(x) = 3x + 2$ says the machine multiplies by 3 and then adds 2.

$f(0) = 2, f(4) = 14, f(-1) = -1$

A table of values collects several inputs at once, which is exactly what you need before drawing a graph:

x	-2	-1	0	1	2	3
$3x + 2$	-4	-1	2	5	8	11

SEQUENCE OR FUNCTION?

A sequence takes only whole-number positions $n = 1, 2, 3, \dots$ — its graph is a row of dots. A function like $f(x) = 3x + 2$ accepts any x , so its graph is an unbroken line.

02

Worked examples

EXAMPLE 1 Find the n th term of 7, 12, 17, 22, ... and use it to find the 30th term.

1

$d = 5$
Each term is 5 more than the one before.

2

$5n$ gives 5, 10, 15, ...
Every term is 2 too small.

3

n th term $= 5n + 2$
Check $n = 3$: $15 + 2 = 17$ ✓

4

$a_{30} = 150 + 2 = 152$

ANSWER

$5n + 2$, 30th term 152

EXAMPLE 2 Find the n th term of 4, 7, 12, 19, 28, ...

① first differences 3, 5, 7, 9
Not constant, so not linear.

② second differences 2, 2, 2
Constant — quadratic, with n^2 inside.

③ n^2 gives 1, 4, 9, 16, 25
Compare term by term.

④ difference is 3 every time

⑤ n th term = $n^2 + 3$
Check $n = 5$: $25 + 3 = 28$ ✓

ANSWER

$$n^2 + 3$$

EXAMPLE 3 If $f(x) = 2x - 5$, find $f(3)$, $f(-2)$ and the x for which $f(x) = 11$.

① $f(3) = 6 - 5 = 1$
Substitute.

② $f(-2) = -4 - 5 = -9$

③ $2x - 5 = 11$
Now run the machine backwards.

④ $2x = 16$, $x = 8$

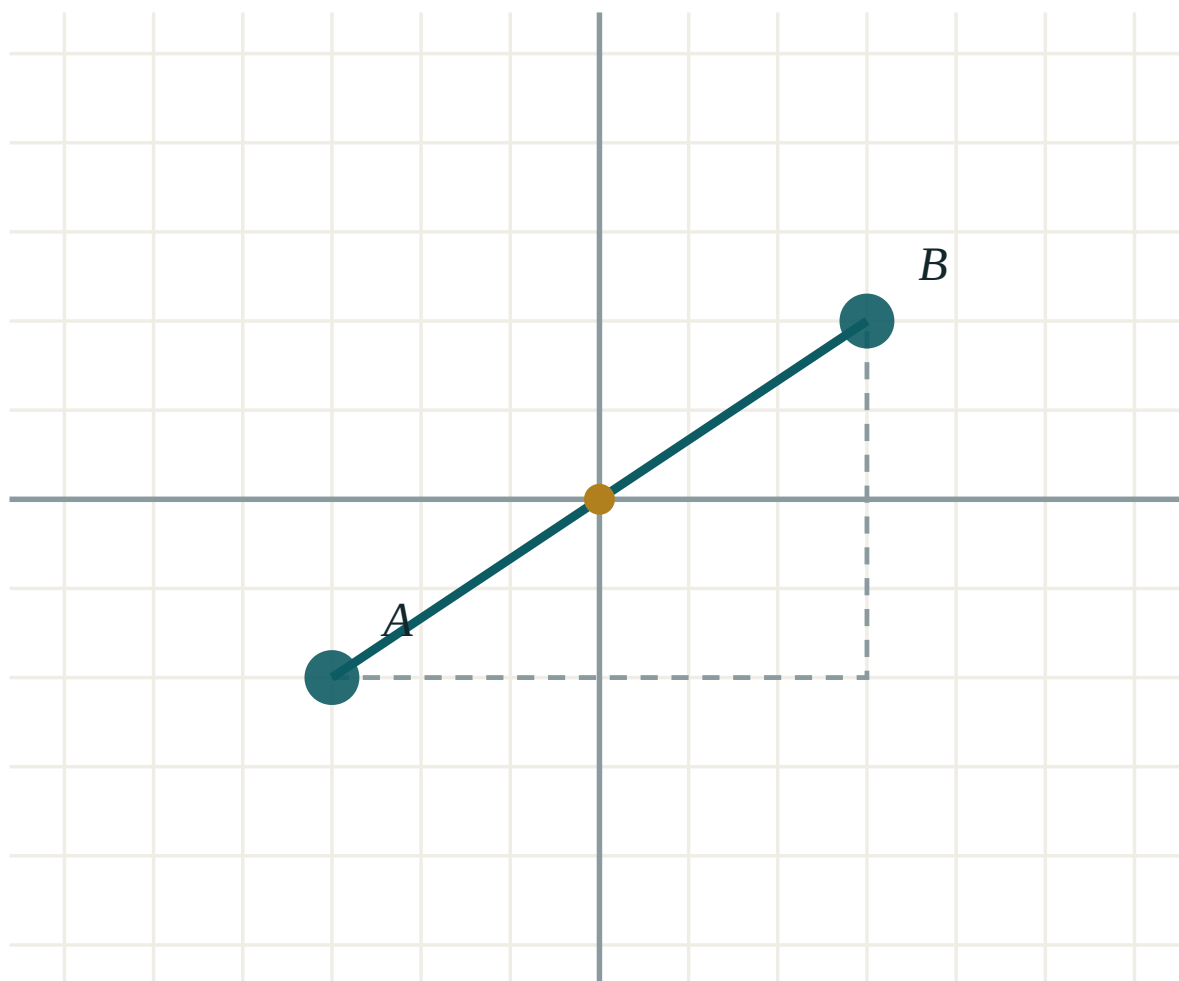
ANSWER

$$f(3) = 1, f(-2) = -9, x = 8$$

03 Interactive model

Drag the points to see how the common difference controls the steepness of a linear pattern.

A sequence plotted as points



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член (последовательности)
Term-to-term rule	Hadlar orasidagi qoida	Рекуррентное правило
Position-to-term rule	O'rin bo'yicha qoida	Формула n-го члена
nth term	n-had	n-й член
Common difference	Ayirma	Разность
Linear sequence	Chiziqli ketma-ketlik	Линейная последовательность
Function	Funksiya	Функция
Input / output	Kirish / chiqish	Вход / выход
Table of values	Qiyimatlar jadvali	Таблица значений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The n th term of 4, 9, 14, 19, ... is:

The 20th term of $3n - 2$ is:

A constant second difference means the sequence is:

If $f(x) = 5 - 2x$ then $f(-3)$ is:

-1

11

-11

1

Which sequence has n th term $n^2 + 1$?

$1, 4, 9, 16$

$2, 5, 10, 17$

$2, 4, 6, 8$

$1, 3, 5, 7$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Write the next two terms of 3, 7, 11, 15, ...

ANSWER 19, 23

2. Find the n th term of 2, 4, 6, 8, ...

ANSWER $2n$

3. Find the n th term of 5, 10, 15, 20, ...

ANSWER $5n$

4. Find the 10th term of $4n - 1$.

ANSWER 39

5. If $f(x) = x + 7$, find $f(3)$.

ANSWER 10

6. If $f(x) = 3x$, find $f(-2)$.

ANSWER -6

7. Write the first four terms of n^2 .

ANSWER 1, 4, 9, 16

Medium · the standard question

7 PROBLEMS

1. Find the n th term of 6, 11, 16, 21, ...

ANSWER $5n + 1$

2. Find the n th term of 20, 17, 14, 11, ...

ANSWER $23 - 3n$

3. Find the 50th term of $7n - 3$.

ANSWER 347

4. Find the n th term of 2, 5, 10, 17, 26, ...

ANSWER $n^2 + 1$

5. If $f(x) = 4x - 3$, solve $f(x) = 17$.

ANSWER $x = 5$

6. Make a table of $2x + 1$ for $x = -2, -1, 0, 1, 2$.

ANSWER $-3, -1, 1, 3, 5$

7. Is 61 a term of $4n + 1$? Justify.

ANSWER Yes — $4n = 60$, $n = 15$, a whole number.

Hard · stretch and reason

7 PROBLEMS

1. Find the n th term of 3, 8, 15, 24, 35, ...

ANSWER $n^2 + 2n$ (i.e. $n(n + 2)$)

2. Is 100 a term of $3n + 2$? Justify.

ANSWER No — $3n = 98$ is not a multiple of 3.

3. The 4th term of a linear sequence is 17 and the 9th is 37. Find the n th term.

ANSWER $d = 4$, n th term $= 4n + 1$

4. Find the n th term of 1, 3, 6, 10, 15, ...

ANSWER $\frac{n(n + 1)}{2}$

5. If $f(x) = x^2 - 4x$, find all x with $f(x) = 0$.

ANSWER $x = 0$ or $x = 4$

6. A sequence starts 2, 6, 12, 20, 30. Find the 20th term.

ANSWER $n(n + 1)$, so 420

7. Explain why a linear sequence can never contain both 7 and 8 if its common difference is 3.

ANSWER Consecutive terms differ by 3, so no two terms differ by 1; every term leaves the same remainder on division by 3, and 7 and 8 do not.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 9. Always check your n th term at $n = 1$ and $n = 3$.

- Find the n th term of 8, 13, 18, 23, ...
- Find the 40th term of $6n - 5$.
- Find the n th term of 3, 6, 11, 18, 27, ...
- If $f(x) = 5x - 4$, find $f(0)$, $f(3)$ and the x with $f(x) = 26$.

5. *Make a table of $3 - 2x$ for $x = -2 \dots 2$.*

6. *Is 73 a term of $5n + 3$? Explain.*

Revision · Graphs, gradient and intercept

Cambridge insert: reading a straight line — and reading a story off one.

LESSON 94–96

3 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 10.1–10.4

LESSON CLOCK

18'

36'

30'

24'

12'

Sketch three lines from memory	18 min
Gradient and intercept	36 min
Parallel and perpendicular	30 min
Real-life graphs	24 min
Homework	12 min

LEARNING OBJECTIVES

- Find the gradient and y-intercept of a straight line from its equation or its graph.
- Draw a line from $y = mx + c$ without a full table of values.
- Recognise parallel and perpendicular lines from their gradients.
- Interpret real-life graphs, including distance–time and conversion graphs.

TEXTBOOK

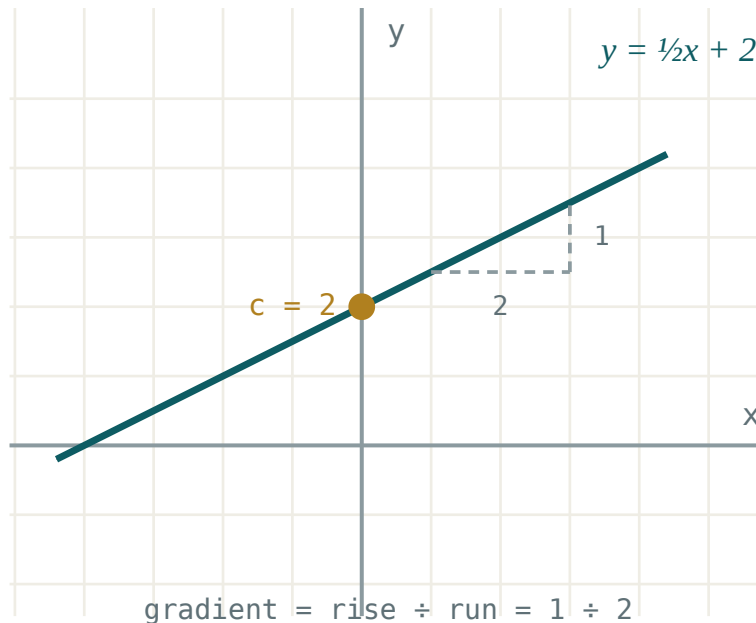
National	Annual revision
Cambridge	Learner’s Book pp. 214–234
Workbook	Workbook 10.1– 10.4

01

Explanation

$y = mx + c$, read straight off

$y = mx + c$ — m is the gradient, c is where the line cuts the y-axis



The gradient is rise ÷ run; c is the height at which the line crosses the y -axis.

$y = 2x - 3$ climbs 2 for every 1 across and passes through $(0, -3)$. That is enough to draw it: mark $(0, -3)$, then step 1 right and 2 up, and join.

GRADIENT FROM TWO POINTS

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Take the differences in the **same order** top and bottom, or the sign comes out wrong.

A positive gradient rises left to right, a negative one falls, and a gradient of 0 is a horizontal line $y = c$. A vertical line $x = k$ has **no** gradient — the run is zero and you cannot divide by it.

Parallel and perpendicular

RELATIONSHIP	GRADIENTS	EXAMPLE
parallel	$m_1 = m_2$	$y = 3x + 1 \parallel y = 3x - 4$
perpendicular	$m_1 \cdot m_2 = -1$	$y = 3x + 1 \perp y = -\frac{1}{3}x + 2$

To find the perpendicular gradient: turn the fraction upside down and change its sign.

For $m = \frac{2}{5}$ the perpendicular gradient is $-\frac{5}{2}$.

An equation given as $2x + 3y = 12$ hides its gradient. Rearrange into $y = mx + c$ form first:

$$3y = -2x + 12 \implies y = -\frac{2}{3}x + 4$$

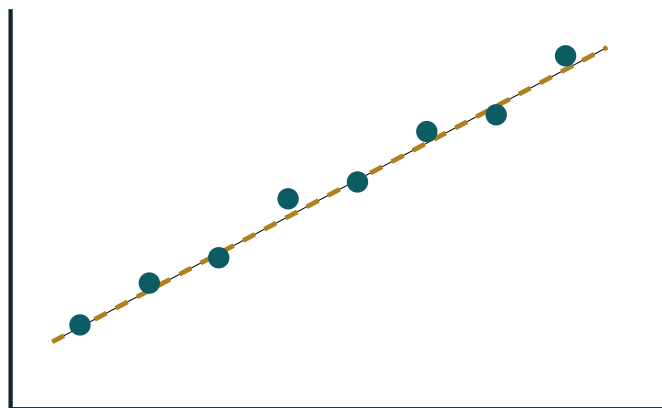
Gradient $-\frac{2}{3}$, intercept 4.

Graphs that tell a story

On a **distance–time** graph the gradient *is* the speed:

- a steep line — moving fast;
- a gentle line — moving slowly;
- a horizontal line — stopped;
- a line coming back down — returning towards the start.

Always read the axes before the shape. The same rising line means “getting faster” on a speed–time graph but “moving at a steady speed” on a distance–time graph.



positive correlation

A scatter graph shows the relationship between two quantities; a line of best fit summarises the trend.

On a **scatter graph** the trend can be positive, negative or absent. A line of best fit passes through the middle of the cloud with roughly as many points above as below — and it is only trustworthy **inside** the range of the data.

CORRELATION IS NOT CAUSE

Ice-cream sales and swimming accidents rise together. Neither causes the other; hot weather causes both.

02 Worked examples**EXAMPLE 1** Find the gradient and y-intercept of $4x + 2y = 10$, and state a parallel line.

① $2y = -4x + 10$

Get the y term alone.

② $y = -2x + 5$

Divide everything by 2.

③ $m = -2, c = 5$

④ $y = -2x + 1$

Any line with the same gradient and a different intercept is parallel.

ANSWER

$$m = -2, c = 5$$

EXAMPLE 2 Find the equation of the line through $(1, 5)$ and $(4, 14)$.

$$\textcircled{1} \quad m = \frac{14 - 5}{4 - 1} = \frac{9}{3} = 3$$

Same order top and bottom.

$$\textcircled{2} \quad y = 3x + c$$

$$\textcircled{3} \quad 5 = 3 \cdot 1 + c$$

Substitute one of the points.

$$\textcircled{4} \quad c = 2$$

$$\textcircled{5} \quad y = 3x + 2$$

Check with $(4, 14)$: $12 + 2 = 14$ ✓

ANSWER

$$y = 3x + 2$$

EXAMPLE**3**

A car travels 120 km in 1.5 h, stops for 0.5 h, then returns in 2 h. Describe the distance–time graph.

① $0 \rightarrow 1.5 \text{ h}$

A straight rise to 120 km; gradient 80 km/h .

② $1.5 \rightarrow 2 \text{ h}$

Horizontal at 120 km — stopped.

③ $2 \rightarrow 4 \text{ h}$

Falls back to 0; gradient -60 km/h , so a speed of 60 km/h homeward.

④ Total journey 4 h, 240 km travelled.

The **displacement** at the end is 0.

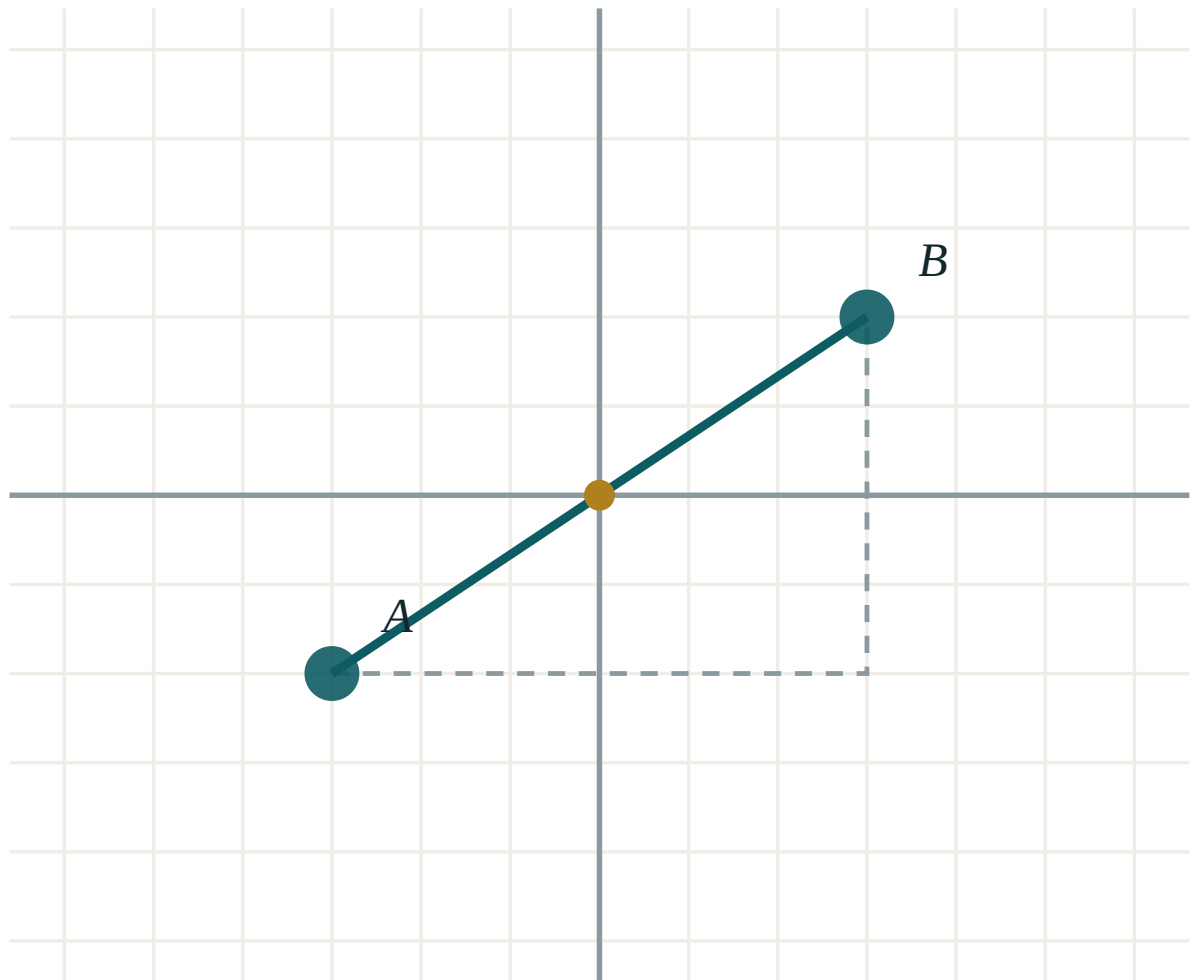
ANSWER

Rise (80 km/h), flat (stopped), fall (60 km/h)

03 Interactive model

Drag the two points and read the gradient off the change in x and the change in y.

Gradient between two points



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Gradient (slope)	Burchak koeffitsienti	Угловой коэффициент
y-intercept	Ordinatalar o'qi bilan kesishuv	Точка пересечения с осью Oy
x-intercept	Abssissalar o'qi bilan kesishuv	Точка пересечения с осью Ox
Rise over run	Ko'tarilish / siljish	Приращение по y / по x
Parallel lines	Parallel to'g'ri chiziqlar	Параллельные прямые
Perpendicular lines	Perpendikulyar to'g'ri chiziqlar	Перпендикулярные прямые
Linear equation	Chiziqli tenglama	Линейное уравнение
Distance–time graph	Masofa–vaqt grafigi	График «путь — время»
Rate of change	O'zgarish tezligi	Скорость изменения
Conversion graph	Almashtirish grafigi	График перевода единиц

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The gradient of $y = 5 - 3x$ is:

A line perpendicular to $y = 4x - 1$ has gradient:

The line through $(0, 3)$ and $(2, 9)$ is:

On a distance–time graph a horizontal segment means:

constant speed

stopped

going backwards

accelerating

Rearranged, $3x - y = 7$ is:

$$y = 3x - 7$$

$$y = 7 - 3x$$

$$y = 3x + 7$$

$$y = -3x - 7$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the gradient and intercept of $y = 4x - 1$.

ANSWER $m = 4, c = -1$

2. State the gradient and intercept of $y = -x + 6$.

ANSWER $m = -1, c = 6$

3. Find the gradient through $(0, 0)$ and $(2, 6)$.

ANSWER 3

4. Does $(2, 7)$ lie on $y = 3x + 1$?

ANSWER Yes — $6 + 1 = 7$

5. Where does $y = 2x - 8$ cut the y -axis?

ANSWER $(0, -8)$

6. Write a line parallel to $y = 5x + 2$.

ANSWER $y = 5x - 3$ (any different intercept)

7. What is the gradient of $y = 7$?

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Rearrange $2x + y = 9$ into $y = mx + c$.

ANSWER $y = -2x + 9$

2. Find the gradient through $(1, 2)$ and $(5, 14)$.

ANSWER 3

3. Find the equation through $(0, -4)$ with gradient 2.

ANSWER $y = 2x - 4$

4. Where does $y = 3x - 12$ cut the x -axis?

ANSWER $(4, 0)$

5. Write a line perpendicular to $y = 2x + 1$ through the origin.

ANSWER $y = -\frac{1}{2}x$

6. A walker covers 6 km in 1.5 h. Find the gradient of the distance–time graph.

ANSWER 4 km/h

7. Rearrange $5x - 2y = 8$ and state the gradient.

ANSWER $y = \frac{5}{2}x - 4, m = \frac{5}{2}$

Hard · stretch and reason

7 PROBLEMS

1. Find the equation through $(2, 1)$ and $(6, -7)$.

ANSWER $y = -2x + 5$

2. Are $3x + 2y = 6$ and $y = \frac{2}{3}x - 1$ perpendicular?

ANSWER Yes — gradients $-\frac{3}{2}$ and $\frac{2}{3}$ multiply to -1 .

3. Find where $y = 2x + 1$ meets $y = 8 - x$.

ANSWER $(\frac{7}{3}, \frac{17}{3})$

4. A line has gradient 3 and passes through $(-2, 1)$. Find its x -intercept.

ANSWER $y = 3x + 7$, so $(-\frac{7}{3}, 0)$

5. The line $y = mx + 4$ passes through $(3, -2)$. Find m .

ANSWER $m = -2$

6. Show that $(1, 3)$, $(3, 7)$ and $(6, 13)$ are collinear.

ANSWER All three pairs give gradient 2, and each point satisfies $y = 2x + 1$.

7. A taxi charges 5000 so‘m plus 2000 so‘m per km. Write the cost as a function of distance and interpret m and c .

ANSWER $C = 2000d + 5000$ — m is the price per km, c the fixed hire charge.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 10. Draw axes from -6 to 6 unless told otherwise.

- State the gradient and intercept of $y = -3x + 5$, and draw it.
- Rearrange $4x + 2y = 10$ and state its gradient.
- Find the equation of the line through $(0, 2)$ and $(4, 10)$.

4. Write a line parallel to, and a line perpendicular to, $y = \frac{1}{2}x + 3$.
5. A cyclist rides 24 km in 2 h, rests 0.5 h, then rides 12 km in 1 h. Sketch the distance–time graph.
6. In one sentence, explain why correlation does not prove cause.

Revision · Ratio, proportion and rates

Cambridge insert: sharing, scaling and comparing — the quiet workhorse of every exam paper.

LESSON 97–98

2 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 11.1–11.2

LESSON CLOCK

12'

20'

20'

20'

8'

Simplify five ratios	12 min
Sharing in a ratio	20 min
Direct proportion	20 min
Inverse proportion and rates	20 min
Homework	8 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use direct proportion, including the unitary method.
- Recognise and use inverse proportion.
- Work with rates, including speed and best value.

TEXTBOOK

National	Annual revision
Cambridge	Learner’s Book pp. 238–252
Workbook	Workbook 11.1–11.2

01

Explanation

Ratio and sharing

A ratio compares parts. Simplify it exactly as you would a fraction, by dividing every term by the same number:

$18 : 24 : 30 = 3 : 4 : 5$

SHARING IN A RATIO

1. Add the parts — that is how many equal shares there are.
2. Divide the total by the number of parts to get **one part**.
3. Multiply back up for each share.

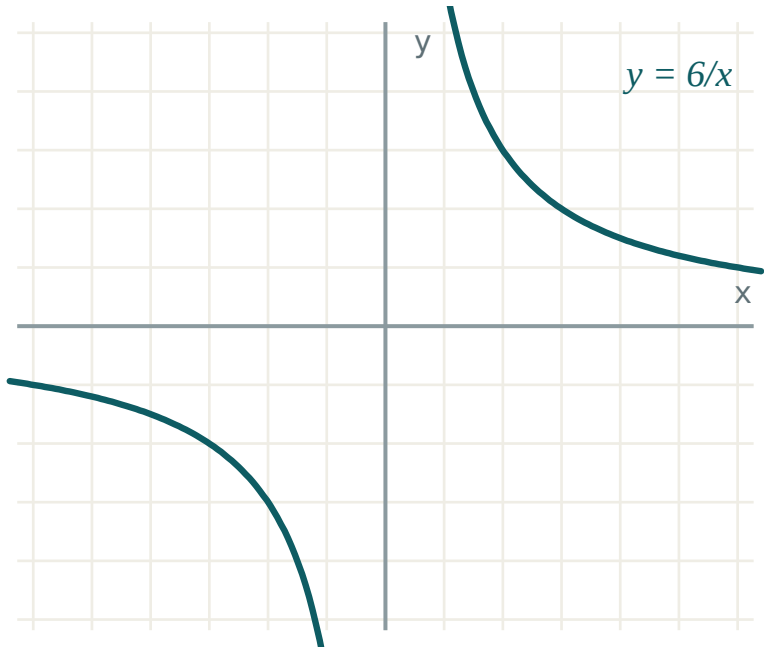
Share 600 000 so’m in the ratio 2 : 3 : 5: there are 10 parts, so one part is 60 000, giving 120 000, 180 000 and 300 000. Check the sum — it must return the original total.

RATIO ≠ FRACTION

In 2 : 3 the first quantity is $\frac{2}{5}$ of the total, not $\frac{2}{3}$. $\frac{2}{3}$ is how it compares with the *other part*.

Direct and inverse proportion

	DIRECT	INVERSE
In words	double one, double the other	double one, halve the other
Rule	$y = kx$	$xy = k$
Graph	straight line through $(0, 0)$	a hyperbola
Example	cost of petrol vs litres	workers vs days to finish



Inverse proportion: as one quantity grows the other shrinks, and their product stays the same.

The **unitary method** settles almost every direct-proportion question: find the value of one, then multiply. If 7 books cost 84 000 so'm, one costs 12 000, so 11 cost 132 000.

For inverse proportion, find the **product** instead. If 4 workers need 18 days, the job is $4 \cdot 18 = 72$ worker-days; 6 workers need $72 \div 6 = 12$ days.

Rates and best value

A **rate** compares two different units — km/h, so'm per kg, litres per 100 km.

$$speed = \frac{\text{distance}}{\text{time}}, \text{ distance} = speed \cdot time, time = \frac{\text{distance}}{speed}$$

For **best value**, reduce every offer to the same unit and compare:

PACK	PRICE	PRICE PER KG
2 kg	36 000	18 000
3 kg	51 000	17 000
5 kg	87 500	17 500

The 3 kg pack is the best value — bigger is not automatically cheaper, which is exactly what the question is testing.

UNITS FIRST

45 minutes is 0.75 hours, not 0.45. Convert before dividing, or the speed is nonsense.

02

Worked examples

EXAMPLE 1 Share 84 sweets between A, B and C in the ratio 3 : 4 : 5.

- 1

$3 + 4 + 5 = 12$
Twelve equal parts.
- 2

$84 \div 12 = 7$
One part.
- 3

$A = 21, B = 28, C = 35$
Multiply back up.
- 4

$21 + 28 + 35 = 84$
Check ✓

ANSWER
21, 28, 35

EXAMPLE 2 5 workers build a wall in 12 days. How long would 8 workers take?

- ① More workers, fewer days — **inverse** proportion.

Do not multiply by $\frac{8}{5}$.

- ② $5 \cdot 12 = 60$ worker-days

The size of the job.

- ③ $60 \div 8 = 7.5$

- ④ $7\frac{1}{2}$ days.

Fewer than 12, as expected.

ANSWER

7.5 days

EXAMPLE 3 A train covers 210 km in 2 h 30 min. Find its average speed, and how far it goes in 40 minutes.

- ① $2\text{ h } 30\text{ min} = 2.5\text{ h}$

Convert first.

- ② $v = \frac{210}{2.5} = 84\text{ km/h}$

- ③ $40\text{ min} = \frac{2}{3}\text{ h}$

- ④ $84 \cdot \frac{2}{3} = 56\text{ km}$

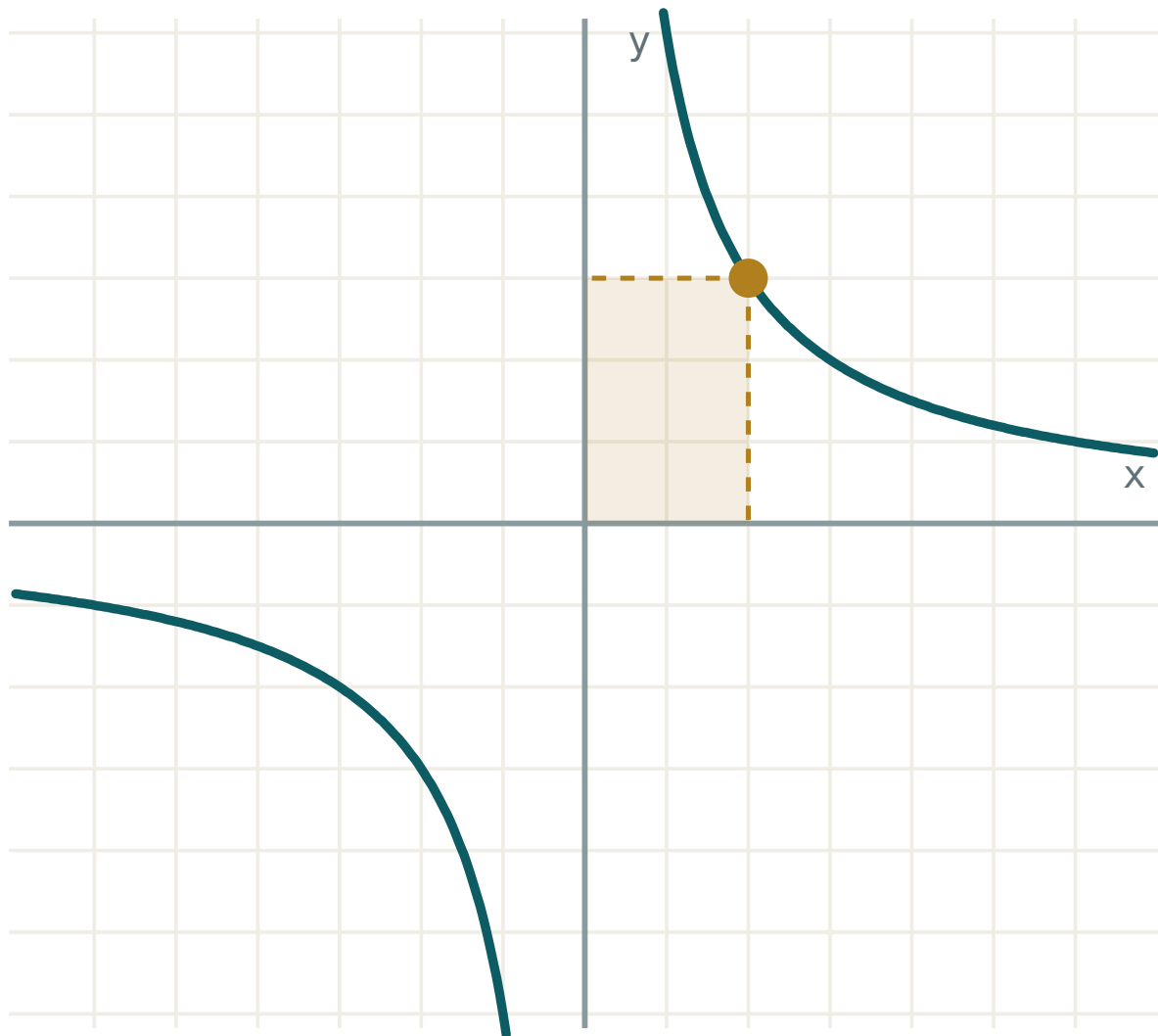
ANSWER

84 km/h; 56 km

03 Interactive model

Slide the value and watch the product stay constant — that is what inverse proportion means.

The graph of $y = k/x$



k 6

x 2

$y = k/x$ 3

$x \cdot y$ 6

quadrants **I** and **III**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Proportion	Proporsiya	Пропорция
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Unitary method	Birlikka keltirish usuli	Метод приведения к единице
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Rate	Tezlik (o'lchov)	Скорость (величина)
Best value	Eng foydali narx	Наиболее выгодная покупка
Share in a ratio	Nisbatda taqsimlash	Разделить в отношении
Constant of proportionality	Proporsionallik koeffitsienti	Коэффициент пропорциональности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

12 : 18 in its simplest form is:

6 : 9

2 : 3

3 : 2

4 : 6

90 shared in the ratio 2 : 3 gives:

30 and 60

36 and 54

45 and 45

20 and 70

6 pipes fill a tank in 10 h. Four pipes take:

15 h

6.7 h

20 h

8 h

3 kg costs 45 000. What do 7 kg cost?

90 000

105 000

115 000

135 000

150 km in 1 h 15 min is a speed of:

100 km/h

120 km/h

125 km/h

115 km/h

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $20 : 35$.

ANSWER $4 : 7$

2. Share 60 in the ratio $1 : 2 : 3$.

ANSWER 10, 20, 30

3. 4 pens cost 20 000 so'm. Find the cost of 7.

ANSWER 35 000 so'm

4. A car travels 180 km in 3 h. Find its speed.

ANSWER 60 km/h

5. Simplify $0.5 : 2$.

ANSWER $1 : 4$

6. If $y = 4x$, find y when $x = 7$.

ANSWER 28

7. 2 kg of rice costs 24 000. Find the price per kg.

ANSWER 12 000 so'm

Medium · the standard question

7 PROBLEMS

1. Share 720 000 so‘m in the ratio 4 : 5 : 7.

ANSWER 180 000, 225 000, 315 000

2. 9 workers finish in 20 days. How long for 12 workers?

ANSWER 15 days

3. 5 books cost 87 500. Find the cost of 8.

ANSWER 140 000 so‘m

4. A journey of 240 km takes 3 h 20 min. Find the speed.

ANSWER 72 km/h

5. y is directly proportional to x , and $y = 18$ when $x = 6$. Find y when $x = 11$.

ANSWER $k = 3, y = 33$

6. Which is better value: 4 kg for 62 000 or 6 kg for 90 000?

ANSWER 6 kg — 15 000/kg against 15 500/kg

7. The ratio of boys to girls is 5 : 4 and there are 20 boys. Find the class size.

ANSWER 36

Hard · stretch and reason

7 PROBLEMS

1. *A and B share money in the ratio 3 : 7. B gets 60 000 more than A. Find the total.*

ANSWER 150 000 so‘m

2. *y is inversely proportional to x, and y = 8 when x = 3. Find y when x = 12.*

ANSWER $k = 24$, $y = 2$

3. *Three people share 500 000 in the ratio 2 : 3 : 5. How much more does the last get than the first?*

ANSWER 150 000 so‘m

4. *A car does 90 km at 60 km/h and 90 km at 90 km/h. Find the average speed for the whole trip.*

ANSWER 72 km/h — $180 \div 2.5$, not the mean of 60 and 90

5. *The ratio a : b is 2 : 3 and b : c is 4 : 5. Find a : b : c.*

ANSWER 8 : 12 : 15

6. *6 machines make 480 parts in 4 h. How many parts do 9 machines make in 5 h?*

ANSWER 900

7. *Explain why doubling the workers does not always halve the time in practice.*

ANSWER The model assumes every worker is equally productive and never gets in another's way; real jobs have tasks that cannot be split and workers who must wait for each other.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 11. Say “direct” or “inverse” before you calculate.

1. Simplify 45 : 60 : 75.
2. Share 480 000 so‘m in the ratio 3 : 5.
3. 7 kg costs 91 000. Find the cost of 12 kg.

4. *8 workers take 15 days. How long do 10 workers take?*
5. *A bus covers 156 km in 2 h 36 min. Find its speed.*
6. *Which is better value: 3 for 21 000 or 5 for 34 000?*

Revision · Probability

Cambridge insert: putting a number on “how likely”, and combining two events.

LESSON 99–100

2 × 40 MIN

ALGEBRA 8 – REVISION

STAGE 9 · 12.1–12.4

LESSON CLOCK



Order five events by likelihood 12 min

The basic formula 20 min

Complement and mutually exclusive events 20 min

Two events 20 min

Homework 8 min

LEARNING OBJECTIVES

- Calculate the probability of an event from equally likely outcomes.
- Use the fact that probabilities of an event and its complement add to 1.
- Use a table or tree diagram for two events.
- Compare experimental with theoretical probability.

TEXTBOOK

National	Annual revision
Cambridge	Learner's Book pp. 255–278
Workbook	Workbook 12.1– 12.4

01 Explanation

Putting a number on chance

FOR EQUALLY LIKELY OUTCOMES

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Every probability lies between 0 (impossible) and 1 (certain).

One die: $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$, $P(7) = 0$, $P(\text{less than } 7) = 1$. Give the answer as a fraction in its simplest form unless a decimal or percentage is asked for.

The word “equally likely” is doing real work. A drawing pin can land point-up or point-down, but not with probability $\frac{1}{2}$ each — only an experiment can tell you.

Complement, and events that exclude each other

$$P(\text{not } A) = 1 - P(A)$$

If the probability of rain is 0.3 , the probability of no rain is 0.7 . This one line turns many “at least one” questions into a single subtraction — exactly as in the counting lesson.

MUTUALLY EXCLUSIVE EVENTS

If A and B cannot happen together, $P(A \text{ or } B) = P(A) + P(B)$.

A die shows a 2 or a 5: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

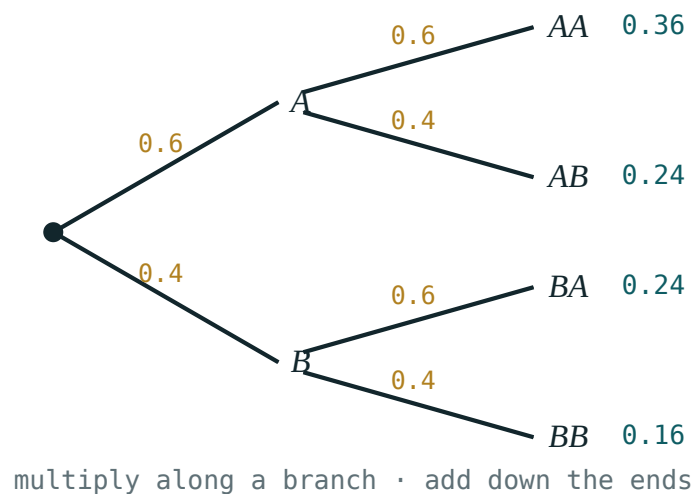
ONLY WHEN THEY EXCLUDE EACH OTHER

“Even” and “greater than 3” are not mutually exclusive — 4 and 6 belong to both — so adding the probabilities double-counts them.

Two events, and what experiments show

For two **independent** events, multiply along the branches of a tree:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$



Multiply along a branch to get the probability of that path; add the paths that satisfy the event.

Two coins: $P(\text{two heads}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. For “exactly one head” there are two successful paths, HT and TH, so add: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

	THEORETICAL	EXPERIMENTAL
Comes from	counting outcomes	actually doing the trials
Formula	$\frac{\text{favourable}}{\text{total}}$	$\frac{\text{times it happened}}{\text{times tried}}$
Behaviour	fixed	settles towards the theoretical value as trials increase

Expected frequency = probability × number of trials. In 300 throws of a die you expect about $300 \cdot \frac{1}{6} = 50$ sixes — “about”, never exactly.

02

Worked examples

EXAMPLE
1

A bag holds 5 red, 3 blue and 4 green counters. One is taken at random. Find $P(\text{red})$ and $P(\text{not green})$.

1

$5 + 3 + 4 = 12$
Total outcomes.

2

$P(\text{red}) = \frac{5}{12}$

3

$P(\text{green}) = \frac{4}{12} = \frac{1}{3}$

4

$P(\text{not green}) = 1 - \frac{1}{3} = \frac{2}{3}$
Use the complement.

ANSWER

$P(\text{red}) = \frac{5}{12}, P(\text{not green}) = \frac{2}{3}$

EXAMPLE 2 Two dice are thrown. Find the probability that the total is 9.

- ① 36 equally likely outcomes
A 6 by 6 table.

- ② (3,6), (4,5), (5,4), (6,3)
Four ways to make 9.

- ③ $P = \frac{4}{36} = \frac{1}{9}$

ANSWER

$$\frac{1}{9}$$

EXAMPLE 3 The probability that a bus is late is 0.2. Find the probability that it is late on both of two days, and that it is late on exactly one.

- ① $P(\text{late, late}) = 0.2 \cdot 0.2 = 0.04$
Independent days — multiply.

- ② $P(\text{late, on time}) = 0.2 \cdot 0.8 = 0.16$
First branch.

- ③ $P(\text{on time, late}) = 0.8 \cdot 0.2 = 0.16$
Second branch.

- ④ $0.16 + 0.16 = 0.32$
Add the two successful paths.

ANSWER

0.04 and 0.32

03 Interactive model

Work through one probability question at a time; ask for the answer before revealing each step.

Probability, step by step

A card numbered 1 to 20 is drawn. Find $P(\text{multiple of } 3)$.

① 3, 6, 9, 12, 15, 18

List
the
favourable
outcomes.

② 6

favourable
out
of

③ $\frac{6}{20} = \frac{3}{10}$

Simplify.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Event	Hodisa	Событие
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход
Complement	Qarama-qarshi hodisa	Противоположное событие
Mutually exclusive	Birgalikda bo'lmaydigan	Несовместные
Independent events	Bog'liq bo'lmagan hodisalar	Независимые события
Tree diagram	Daraxt diagrammasi	Дерево вероятностей
Experimental probability	Tajribaviy ehtimollik	Опытная (частотная) вероятность
Expected frequency	Kutilayotgan chastota	Ожидаемая частота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A die is thrown. $P(\text{more than } 4)$ is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

If $P(A) = 0.42$ then $P(\text{not } A)$ is:

$$0.42$$

$$0.58$$

$$0.68$$

$$1.42$$

Two coins are tossed. $P(\text{exactly one head})$ is:

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{3}{4}$$

A spinner has $P(\text{blue}) = 0.25$. In 80 spins expect about:

16

20

25

40

Which pair is NOT mutually exclusive for one die?

"2" and "5"

"even" and "odd"

"even" and "more than 3"

"1" and "6"

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A die is thrown. Find $P(3)$.

ANSWER $\frac{1}{6}$

2. A die is thrown. Find $P(\text{odd})$.

ANSWER $\frac{1}{2}$

3. A bag has 3 red and 7 blue. Find $P(\text{red})$.

ANSWER $\frac{3}{10}$

4. If $P(\text{win}) = 0.4$, find $P(\text{not win})$.

ANSWER 0.6

5. A coin is tossed twice. Find $P(\text{two tails})$.

ANSWER $\frac{1}{4}$

6. A card is drawn from 1–10. Find $P(\text{more than 7})$.

ANSWER $\frac{3}{10}$

7. A spinner has $P(\text{red}) = 0.5$. In 60 spins, expect how many reds?

ANSWER 30

Medium · the standard question

7 PROBLEMS

1. *Two dice are thrown. Find $P(\text{total } 7)$.*

ANSWER $\frac{1}{6}$

2. *A bag has 4 red, 5 blue, 6 green. Find $P(\text{not blue})$.*

ANSWER $\frac{2}{3}$

3. *$P(\text{late}) = 0.15$. Find $P(\text{late on both of two days})$.*

ANSWER 0.0225

4. *A coin is tossed three times. Find $P(\text{exactly two heads})$.*

ANSWER $\frac{3}{8}$

5. *A card is drawn from 1–20. Find $P(\text{prime})$.*

ANSWER $\frac{8}{20} = \frac{2}{5}$

6. *$P(A) = 0.3$, $P(B) = 0.5$ and A, B are mutually exclusive. Find $P(A \text{ or } B)$.*

ANSWER 0.8

7. *A spinner has $P(\text{green}) = 0.35$. In 400 spins, expect how many greens?*

ANSWER 140

Hard · stretch and reason**7 PROBLEMS**

1. A bag has 5 red and 4 blue. Two are taken without replacement. Find $P(\text{both red})$.

ANSWER $\frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$

2. Two dice are thrown. Find $P(\text{at least one 6})$.

ANSWER $\frac{11}{36}$

3. $P(\text{rain}) = 0.4$ on each of three days. Find $P(\text{no rain at all})$.

ANSWER $0.6^3 = 0.216$

4. $P(\text{rain}) = 0.4$ on each of three days. Find $P(\text{at least one wet day})$.

ANSWER $1 - 0.216 = 0.784$

5. A bag has 6 red and 4 blue. One is taken and replaced, then another. Find $P(\text{different colours})$.

ANSWER $2 \cdot 0.6 \cdot 0.4 = 0.48$

6. A die is thrown 120 times and gives 32 sixes. Compare experimental and theoretical probability.

ANSWER $\frac{32}{120} \approx 0.27$ against $\frac{1}{6} \approx 0.17$ — noticeably high; either chance, or the die is biased

7. Explain why $P(A \text{ or } B) = P(A) + P(B)$ can fail.

ANSWER If A and B can happen together, the outcomes in both are counted twice; the correct rule subtracts $P(A \text{ and } B)$.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 12. Simplify every fraction.

- A bag has 7 red and 5 blue. Find $P(\text{blue})$ and $P(\text{not blue})$.
- Two dice are thrown. Find $P(\text{total 5})$.

3. *A coin is tossed three times. Find $P(\text{at least one tail})$.*
4. *$P(\text{faulty}) = 0.05$. In 600 items, how many faulty ones are expected?*
5. *A card is drawn from 1–15. Find $P(\text{multiple of 4})$.*
6. Explain the difference between experimental and theoretical probability in two sentences.

Control work 8 · Annual review

The whole of Grade 8 algebra in one paper — and the four habits that decide the grade.

LESSON 101–102

2 × 40 MIN

ALGEBRA 8 – ANNUAL REVIEW

STAGE 9 · UNITS 1–15

LESSON CLOCK

4'

72'

4'

Setting up	4 min
The paper	72 min
Collect in	4 min

LEARNING OBJECTIVES

- Assess the year: fractions, roots, quadratics, inequalities, data and counting.
- Diagnose each lost mark by error type.
- Set out the summer work that follows from the result.

TEXTBOOK

National	Chapters I–IV
Cambridge	Learner’s Book, whole course
Workbook	Workbook, whole course

01

Explanation

Lesson 101 — the paper (40 minutes)

Two variants of ten tasks. Tasks 1–6 are worth 1 mark, tasks 7–10 are worth 2:

- task 1 · simplify an algebraic fraction
- task 2 · add two fractions with different denominators
- task 3 · simplify a square root
- task 4 · solve a quadratic equation
- task 5 · solve a linear inequality and show it on a number line
- task 6 · find the mean, median and mode of a small list
- task 7 · a word problem leading to a quadratic
- task 8 · a system of two inequalities
- task 9 · the n th term or the gradient of a line
- task 10 · a counting or probability question

Marking note: in task 1 the excluded values must be stated; in task 7 the impossible root must be rejected in writing.

Lesson 102 — work on mistakes and the summer plan (40 minutes)

THE FOUR HABITS THAT DECIDE THIS PAPER

1. **Write the domain** before touching an algebraic fraction.
2. **Never divide by an unknown** — factorise, so that $x = 0$ is not lost.
3. **Reverse the inequality sign** whenever you multiply or divide by a negative number.
4. **Check the answer against the question**, not just against your own working — a length cannot be negative, a probability cannot exceed 1.

Each pupil writes the habit number beside every lost mark. The topics that appear twice in that list become the summer plan: five problems a week from that section, no more.

LOOKING AHEAD TO GRADE 9

Everything next year rests on three things from this one: factorising confidently, solving a quadratic without hesitation, and reading a graph. If any of those is shaky, that is where the summer goes.

02 Worked examples

EXAMPLE 1 Find the error: $\frac{x^2 - 9}{x - 3} = x + 3$ for every x

- ① The cancelling is right, but the domain is not.
 $x - 3$ is in the denominator.

- ② $x \neq 3$
 At $x = 3$ the original fraction has no value at all.

- ③ $\frac{x^2 - 9}{x - 3} = x + 3, x \neq 3$
 The restriction is part of the answer.

ANSWER

$$x + 3, x \neq 3$$

EXAMPLE 2 Find the error: $-2x > 6$ so $x > -3$

① Both sides were divided by -2 .
That reverses the order.

② $x < -3$
The sign turns round.

③ $x = -5$: $-2 \cdot (-5) = 10 > 6$ ✓
Test a value to be sure.

ANSWER

$$x < -3$$

03 Interactive model

Show each piece of wrong working, take a vote on the habit that was missing, then reveal.

Which habit was missing?

Claimed: $\frac{x^2 - 4}{x - 2} = x + 2$ for every x

① $x^2 - 4 = (x - 2)(x + 2)$

The
factorising
is
correct.

② The denominator vanishes at

$x = \frac{\quad}{2}$

There
the
original
expression
is
undefined.

③ Write

$x \neq \frac{\quad}{2}$ beside the answer.

A
mark
hangs
on
that
line.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual review	Yillik takrorlash	Годовое повторение
Control work	Nazorat ishi	Контрольная работа
Domain (allowed values)	Aniqlanish sohasi	Область допустимых значений
Discriminant	Diskriminant	Дискриминант
Inequality	Tengsizlik	Неравенство
Modulus	Modul	Модуль
Estimated mean	Taxminiy o'rta	Оценка среднего
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Check by substitution	O'rniga qo'yib tekshirish	Проверка подстановкой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 - 25$ $x - 5$ simplifies to:

$-4x < 20$ gives:

$2x^2 = 10x$ has roots:

A probability comes out as 1.2. This means:

the event is certain

there is an error

the event is very likely

multiply by 100

$\sqrt{50}$ in simplest form is:

$5\sqrt{2}$

$2\sqrt{5}$

$25\sqrt{2}$

$10\sqrt{5}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Simplify $\frac{x^2 - 16}{x + 4}$ and state the excluded value.

ANSWER $x - 4, x \neq -4$

2. **Task 2.** Simplify $\frac{1}{x} + \frac{1}{2x}$

ANSWER $\frac{3}{2x}, x \neq 0$

3. **Task 3.** Simplify $\sqrt{72}$

ANSWER $6\sqrt{2}$

4. **Task 4.** Solve $x^2 - 7x + 10 = 0$

ANSWER $x = 2, x = 5$

5. **Task 5.** Solve $3x - 4 < 11$

ANSWER $x < 5$

6. **Task 6.** Find the mean, median and mode of 4, 8, 4, 6, 8, 4

ANSWER mean 5.67, median 5, mode 4

7. **Task 9.** Find the n th term of 5, 9, 13, 17, ...

ANSWER $4n + 1$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Simplify $\frac{3x + 12}{x^2 - 16}$

ANSWER $\frac{3}{x - 4}, x \neq \pm 4$

2. **Task 2.** Simplify $\frac{2}{x - 1} - \frac{1}{x}$

ANSWER $\frac{x + 1}{x(x - 1)}, x \neq 0, 1$

3. **Task 4.** Solve $2x^2 - 7x + 3 = 0$

ANSWER $x = 3, x = 0.5$

4. **Task 5.** Solve $5 - 2x \geq 1$ and show it on a number line.

ANSWER $x \leq 2$

5. **Task 8.** Solve the system $x + 1 > 0$ and $2x - 6 < 0$.

ANSWER $-1 < x < 3$

6. **Task 9.** Find the gradient of the line through (1, 4) and (5, 16).

ANSWER 3

7. **Task 10.** Two dice are thrown. Find $P(\text{total } 6)$.

ANSWER $\frac{5}{36}$

Hard · stretch and reason

7 PROBLEMS

1. **Task 7.** A rectangle is 4 cm longer than wide and has area 96 cm². Find its sides.

ANSWER $8\text{ cm} \times 12\text{ cm}$ (reject -12)

2. **Task 7, variant 2.** A number and its square add to 56. Find the positive value.

ANSWER 7

3. Solve $|x - 3| < 5$

ANSWER $-2 < x < 8$

4. Solve $x^2 - 5x + 6 > 0$

ANSWER $x < 2$ or $x > 3$

5. A code of 3 digits from 1–5 must contain at least one 5. How many codes?

ANSWER $125 - 64 = 61$

6. Simplify $\frac{x^2 - 5x + 6}{x^2 - 9}$ and state the excluded values.

ANSWER $\frac{x - 2}{x + 3}$, $x \neq \pm 3$

7. One root of $x^2 + bx + 21 = 0$ is 3. Find b and the other root.

ANSWER $b = -10$, other root 7

07 Homework

Summer work — 5 tasks

Not a punishment: twenty minutes a week keeps Grade 9 easy.

1. Re-solve, in full, each task you lost a mark on, with the habit number (1–4) beside it.
2. List the two topics that appear most often in that list — those are your summer topics.
3. Five problems a week from each of those two topics. No more.
4. Practise until you can factorise $x^2 + bx + c$ and solve any quadratic without hesitating.

5. *Practise reading the gradient and intercept off any straight-line graph.*